

Impact Evaluators - Evan Miyazono



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Retroactive public goods funding for the broader Protocol Labs network. Funding the Commons Conference Learn more at <https://fundingthecommons.io/> 0:00 Introduction 1:03 The problem of proportional upside 7:07 A variety of retroactive funding mechanisms 12:36 A network of retroactive funding mechanisms 16:28 Scaling the solution

Introduction

0:02

we'll be kicking off this session with

0:04

the talk by evan miazono

0:06

evan leads the research team at protocol

0:08

labs and his additional interests

0:10

include lock picking and citrus fruits

0:12

today he'll discuss retroactive public

0:14

goods funding for the broader protocol

0:17

labs network

0:19

welcome evan

0:20

thank you for the introduction carlo and

0:22

uh yes i'm going to be talking about

0:25

uh

0:26

what we've been calling impact evaluated

0:28

internally and what

0:31

what purpose we hope they serve what

0:32

we'll be doing with them and

0:36

hopefully tying together some very

0:37

interesting threads that have come up

0:38

over the course of today's conversations

0:42

overall there's a very real problem that

0:44

protocol labs is trying to solve and the

0:47

hope expectation etc is that the

0:49

solution comes in really two parts and

0:52

i'm planning on talking through those

0:53

two parts and then explaining what you

0:55

should expect to see coming out of

0:57

protocol labs in the near future and how

0:58

to engage with us

1:00

trying to collaboratively build

1:01

solutions to those problems

The problem of proportional upside

1:04

in terms of the

1:06

the problem i would say that it's one of

1:08

proportional upside

1:11

you'll notice some similarities between

1:12

this talk and the talk that puja gave

1:14

earlier we're both kind of approaching

1:17

the same

1:18

general problem uh

1:20

from in two different ways two different

1:22

subparts of the same problem and i would

1:24

definitely recommend watching that video

1:26

if you're catching this uh

1:28

uh if you missed that one

1:30

uh and

1:32

overall i would say that

1:34

this particular problem the one of

1:35

proportional upside is one of two very

1:38

related ones

1:40

and it will start with me coming back

1:42

with some

1:44

context of what protocol ads is looking

1:46

at specifically

1:48

we are working to empower and accelerate

1:51

teams outside of protocol ads the

1:53

organization

1:55

uh

1:56

we think that this is a fantastic

1:58

opportunity because the dependency graph

1:59

of crypto projects is very large but

2:02

also very dense

2:04

not all project have projects have

2:05

revenue streams but there are some clear

2:06

dependencies between those that do and

2:08

some don't the ones that don't being

2:10

some of these public goods that we've

2:11

been talking about all day and

2:14

we think that there's a really robust

2:16

opportunity to share wealth where

2:17

incentives are aligned um i.e

2:20

referencing back to bonds talk we think

2:22

that prototopia can robustly be achieved

2:24

in this kind of a network and so we've

2:27

been

2:28

thinking through a lot of

2:29

these types of mechanisms how to connect

2:31

these types of mechanisms uh to help

2:33

people start public good type efforts

2:38

we have a few

2:42

funding mechanisms that

2:44

will let people

2:46

de-risk initially starting on these

2:48

efforts and also um

2:50

how do we empower them to continue doing

2:52

these this is specifically what fuji was

2:54

getting into earlier today

2:56

but

2:57

there's a chance that with grants or

2:59

bounties uh

3:01

you might have this feeling that you're

3:04

barely laying down enough runway to

3:08

cover your current activities i've seen

3:10

for public goods like academic research

3:12

this happened where fantastic labs

3:15

tend to have bad luck because they took

3:17

on a project that was too risky and they

3:19

were unable to fund their lab and had to

3:21

shrink down based on either the whims of

3:24

grant makers or the

3:26

effectiveness capital effectiveness of

3:28

tech transfer offices which ended up

3:29

being fairly capricious and so i don't

3:32

think that this problem is solved well

3:35

at all in academia it's definitely not

3:37

solved particularly well for open source

3:39

and so

3:40

uh

3:41

i think that there's

3:43

definitely something more to be done

3:44

here in addition to

3:46

grants and bounties

3:48

which do serve an important role i would

3:50

say that for funding a network you want

3:52

these two properties to be satisfied

3:56

you want teams to be able to have runway

3:59

to take on things that have high risk

4:01

high reward

4:02

when i think about

4:04

executing on project in space it's very

4:06

easy to see that many people are not

4:09

able to take the kind of risks

4:11

to pursue things that they think are

4:13

valuable

4:15

because they don't have

4:17

the privilege or the resources and so

4:20

being able to provide that some support

4:22

network like thing to let people span

4:24

that gap is very important so they can

4:26

take on things that have higher expected

4:30

outcome uh however it's also very

4:32

important that teams be incentivized to

4:34

do those very impactful things because

4:36

even if you're unable to it may not be

4:38

the rational choice to go and do those

4:41

not that it necessarily needs to be the

4:42

rational choice but it

4:44

never really hurts to have

4:47

the prospect of high payoff in these

4:50

situations

4:52

and so i would claim that puja's talk

4:55

should have convinced you the answer to

4:56

this first question is yes we can use

4:59

things like milestone bounties um and

5:01

grants to give teams runway the second

5:04

question the answer i think is a little

5:05

less clear if you're here you probably

5:07

agree that this is possible um

5:10

italic highlighted some great ways in

5:11

which the crypto community is really

5:13

well suited to public goods funding i

5:15

totally agree you have things like

5:17

computational control of value which

5:19

is harder for groups outside of the

5:21

cryptocurrency space to deal with

5:23

effectively you have strong belief in

5:25

the transformative power of technology

5:27

and you have

5:29

what has historically been a fairly

5:31

exponential growth of value capture and

5:33

wealth and

5:36

harkening back to juan's taco highlight

5:38

that pradoxia looks like

5:41

a place in which there are so many

5:43

low-hanging fruits there are so many

5:44

valuable things um that there aren't

5:47

enough teams to take on all of them and

5:48

you see huge amounts of collaboration in

5:50

the space as opposed to competition and

5:53

i think if you look around in

5:56

web 3 that's really what you see

5:58

and so the question becomes how do we go

6:00

about answering

6:02

this second question with a strong

6:03

affirmative and i unsurprisingly the

6:06

answer that i would give is um

6:09

public goods funding

6:11

specifically retroactive funding

6:13

my claim is that

6:16

if you

6:18

if you want to empower people to take

6:20

risks you need to fund proactively if

6:22

you want to reward

6:24

impact you need to do it retroactively

6:27

if you

6:29

incentivize impact proactively then what

6:30

you're actually doing i think in

6:32

practice is incentivizing the appearance

6:34

of effort not the appearance of impact

6:36

which is an important distinction

6:39

and you can imagine that a combination

6:42

of these funding mechanisms

6:44

could empower teams to not just take on

6:48

risky efforts but also be rewarded

6:50

proportionally for them um this plot on

6:53

the right shows what we would predict

6:55

what a

6:56

hypothetical team without a revenue

6:58

model might be able to gain from some

7:00

initial grant some bounties and reward

7:03

from what we're calling the impact

7:04

evaluator and so

7:06

the first component from the solution

A variety of retroactive funding mechanisms

7:09

of the solution for these impact

7:11

evaluators

7:12

that i would say is a variety of

7:14

retroactive funding mechanisms these are

7:17

things that will sound very familiar

7:19

because most of what

7:21

have been covered so far today

7:23

could be characterized as retroactive

7:26

funding mechanisms

7:28

i'm not going to go into depth i will

7:29

simply say

7:31

that

7:32

uh if you haven't seen them you should

7:34

definitely go and watch owens talking

7:36

earlier today on bitcoin grants and

7:38

quadratic voting you could use quadratic

7:40

voting as an impact evaluator

7:43

uh you could use

7:46

uh

7:48

uh badge holder voting and uh in the

7:52

example of the

7:53

retroactive public goods funding from

7:55

the optimism team uh the

7:59

results oracle is an impact evaluator

8:01

these impact values are just things that

8:03

are after the fact evaluating and

8:05

rewarding impact to teams through

8:07

various mechanisms

8:10

well they're definitely more

8:11

representative slides of retroactive

8:12

public goods funding but i definitely

8:14

had to go for one of carl's sides of the

8:16

hamsters naturally um

8:19

uh in terms of

8:21

s process i unfortunately had to miss

8:23

much of andrea's talk but

8:25

you could additionally use

8:28

things like

8:30

eliciting marginal utility functions for

8:34

determining how to allocate rewards

8:36

retroactively

8:39

it does seem to work far more

8:40

effectively prospectively but

8:43

in terms of developing a wide spectrum

8:46

of these types of mechanisms

8:48

because it's definitely worth

8:49

considering um

8:52

additionally we had an interesting

8:54

experiment protocol labs did um starting

8:56

in the fall of 2019 there was a project

8:58

called source credit that we helped

9:01

bootstrap

9:02

where

9:03

data was pulled from

9:06

github discourse discord

9:09

into some global graph

9:11

the uh

9:13

different contributions were

9:16

collapsed using

9:17

uh

9:19

some

9:20

community-driven subjective agreement on

9:23

how we should or how the community

9:25

should

9:27

quantify different types of

9:28

contributions be able to collapse all of

9:30

this down into a single contribution

9:31

graph run a modified page rank algorithm

9:33

on this and then assign some reputation

9:36

score called cred to individual

9:39

contributors of the project and then

9:40

grain was this

9:42

fungible token that was

9:45

generated and given to contributors in

9:48

proportion to their credit score with

9:49

the idea being that you could

9:51

computationally allocate value

9:54

in accordance to the value that people

9:56

provided to the project

9:59

and the experiment that i mentioned

10:01

about source credit

10:02

is that for about two years protocol

10:05

labs funded source cred through the use

10:07

of source cred those of you who are

10:09

familiar with the retroactive public

10:11

goods funding will recognize this as

10:14

very similar to a project token

10:17

and i'm very interested in what it would

10:19

be like to build

10:22

better and more robust project token

10:23

examples um source code very

10:25

aggressively dog foods tool but they got

10:27

to the point where

10:28

they are effectively independent and for

10:31

quite a period of time

10:33

protocol labs supported only through the

10:35

purchasing of the project token

10:39

in terms of other impact evaluators that

10:42

you might expect to see

10:43

moving forward protocol labs is

10:45

experimenting and prototyping

10:47

a committee quantification impact

10:49

evaluator where

10:51

outcomes are listed

10:53

uh

10:54

rather

10:55

teams participating in the network opt

10:57

in to participate in this process and

11:01

summarize their outcomes and then at the

11:05

on a quarterly cadence some evaluation

11:08

team that consists of people within the

11:09

network will go through and evaluate the

11:12

amount of impact

11:15

not too dissimilar from the badge

11:17

holders

11:18

and

11:19

will determine how much how many

11:21

resources should be allocated

11:23

we also expect that there will be other

11:26

experimental impact value

11:28

evaluators that we generate

11:31

another one could be something looks

11:32

more like a block reward analog

11:36

where instead of

11:38

minting tokens we create some

11:41

general pot of

11:44

assets

11:46

define some deadline some maximum reward

11:48

some kpi

11:50

and

11:52

it's important that we have some sort of

11:53

upward concavity so that as

11:57

teams compete to maximize the kpi or as

12:00

teams try to maximize the kpi they do so

12:02

collaboratively rather than

12:03

competitively so because as

12:06

uh

12:08

teams achieve outcomes the reward is

12:11

strongly nonzero sum and so we can

12:13

generate this like build in this kind of

12:15

non-zero sum

12:16

effect

12:17

into the reward itself and then allocate

12:19

using

12:21

something like uh

12:24

either direct contribution with kpi if

12:25

that's something you could do you could

12:27

also apply something like sharply values

12:29

in order to generate

12:32

a more subjective evaluation

A network of retroactive funding mechanisms

12:36

that's a general set of a bunch of

12:37

different types of impact evaluators

12:39

that you could have

12:41

the second part of the solution to allow

12:43

uh

12:44

projects to be incentivized to be

12:46

incentivized to pursue very risky things

12:49

is to

12:51

acknowledge that not any one of these

12:53

impact evaluators is likely to be

12:55

perfect for all

12:57

possible types of work all possible

12:59

types of efforts all teams

13:02

and so

13:04

the goal will be to

13:07

take these retroactive funding

13:08

mechanisms and develop them into a

13:10

network that leverages the fact that

13:13

protocol labs is building

13:15

around its projects a network

13:18

the

13:19

first thing worth clarifying is that

13:21

when one of these impact evaluators

13:23

quantifies impact what they're doing is

13:27

they are

13:28

choosing some metric for themselves that

13:30

is likely a subjective metric um

13:33

in the case of protocol labs this is

13:35

probably likely to be

13:38

the pii mission generally or it could be

13:40

something

13:42

related to a sub-aspect of it around

13:44

increasing the

13:46

ownership that users have over their own

13:48

data for instance

13:51

and

13:52

each of these impact evaluators is going

13:54

to have to find some way

13:56

through some people to subjectively

13:58

quantify

14:00

the impact it was made against uh

14:03

some goal to reduce this to numbers uh

14:05

in order to motivate that this is

14:08

reasonable and plausible i would

14:10

highlight that this kind of subjective

14:12

evaluation happens all the time in a

14:14

very centralized and permissioned way in

14:17

normal companies through things like

14:18

performance incentives or payments and

14:20

equity where the evaluation is actually

14:22

done later well partially done

14:26

directly immediately through the

14:27

allocation of equity and then later

14:29

through the evaluation of the market on

14:31

that equity

14:33

the other component that's very

14:34

interesting about mixing these together

14:35

is that we should expect

14:37

there will be multiple impact evaluators

14:40

in any given network

14:43

and then we should have multiple funders

14:46

each of their own mission or goals

14:47

participating into

14:49

and funding some impact evaluator

14:51

uh each impact evaluator would have its

14:53

own mechanism and uh

14:57

operation and each one would then reward

14:59

multiple teams and so

15:01

you should have much better coverage of

15:03

all the different types of work that you

15:05

would like to incentivize through these

15:06

and we're actively working through and

15:08

trying to develop

15:10

a mechanism to mix these together well

15:14

it's worth noting that

15:18

the

15:19

goal of this system is

15:22

to incentivize teams to pursue the most

15:24

impactful work they can

15:26

and

15:29

as a result retroactive funding

15:32

needs to be something that people can

15:33

expect and count on

15:35

and so

15:37

while trusting that for instance

15:39

protocol apps would reward a project for

15:41

uh retroactive

15:43

would retroactively reward a project for

15:44

building something that was important i

15:46

think it's far more valuable if

15:48

we can

15:50

together build a culture of retroactive

15:52

public goods funding so that we can

15:57

give teams the confidence that if they

15:59

do something that is very impactful they

16:01

don't have to they priority know that

16:03

the team that they're having an impact

16:04

on

16:05

has an impact evaluator or one of the

16:07

teams they're having an impact on and so

16:08

i think that having events like this

16:10

bringing people together and

16:13

developing new solutions and

16:17

making sure that these solutions are

16:20

widespread and effective will do wonders

16:22

for increasing the amount of public

16:24

goods that are created

16:25

not even in the very long term

Scaling the solution

16:28

and so in terms of how we are going

16:31

about scaling our solution and

16:34

ways in which you can observe and

16:36

participate in our efforts i'd highlight

16:38

first that we are

16:40

setting some very ambitious goals around

16:42

what we want to accomplish within the

16:44

next year in terms of impact evaluators

16:46

we definitely want to have scale we want

16:47

to have allocation of significant

16:49

amounts of funding

16:50

and

16:53

we want to make sure that people

16:55

involved in

16:56

participating on these processes whether

16:58

they are funders or recipients we want

17:01

them to

17:02

effectively right we want them to find

17:05

these effective uh quickly going through

17:07

the last few

17:09

uh notes i will say that we have some

17:11

open roles we are trying to actively

17:13

build and support more of these research

17:14

roles will look more like identifying

17:16

and defining mechanisms startup operator

17:18

we'll look more about operationalizing

17:19

and running these mechanisms

17:21

we're definitely interested in

17:23

cross-pollinating so if you're

17:24

interested in running experiments

17:26

either co-funding impact evaluators are

17:28

asking us to go fund impact developers

17:30

please let us know

17:33

in terms of

17:36

like

17:37

trying to improve these there are

17:38

clearly gaps

17:40

i think there are some very good

17:41

questions about how these can improve

17:43

and

17:44

what we should be doing to make

17:47

any or all of these impact evaluators

17:49

together more effective and i would

17:52

suggest that you

17:53

reach out uh if you have any comments or

17:56

questions on any of these here's my

17:57

contact information and i don't know if

17:59

i have time for questions we ran out of

18:02

time for questions unfortunately but we

18:04

can also ask them retroactively after

18:06

the event in the telegram channel

18:09

and i'd like to thank evan for walking

18:11

us through the impact evaluator which

18:14

sounds like it should hurt but is

18:16

actually a super interesting journey

18:18

through innovative funding mechanisms so

18:20

thanks again

18:30

you